

Database Technologies – Assignment 9

Create appropriate indexes to solve the following problems. Note: use “sales” database to solve below queries.

1. Create an index that will enable a user to pull orders for ‘1990-10-03’ out of the Orders table quickly.

```
W2_99546_Pratiksha>CREATE INDEX idx_odate
-> ON ORDERS(ODATE);
Query OK, 0 rows affected, 1 warning (0.06 sec)
Records: 0 Duplicates: 0 Warnings: 1

W2_99546_Pratiksha>SELECT * FROM ORDERS
-> WHERE ODATE = '1990-10-.3';

+-----+-----+-----+-----+-----+
| onum | amt   | odate   | cnum | snum |
+-----+-----+-----+-----+-----+
| 3001 | 18.69 | 1990-10-03 | 2008 | 1007 |
| 3003 | 767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
+-----+-----+-----+-----+-----+
5 rows in set, 1 warning (0.01 sec)
```

2. If the Orders table has already been created, how can you force the onum field to be unique (assume all current values are unique)?

```
W2_99546_Pratiksha>CREATE UNIQUE INDEX idx_onum
-> ON ORDERS(ONUM);
Query OK, 0 rows affected (0.04 sec)
Records: 0 Duplicates: 0 Warnings: 0

W2_99546_Pratiksha>SHOW INDEX FROM ORDERS;

+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| Table | Non_unique | Key_name | Seq_in_index | Column_name | Collation | Cardinality | Sub_part | Packed | Null | Index_type | Comment | Index_comment | Visible | Expression |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| orders | 0 | idx_onum | 1 | onum | A | 10 | NULL | NULL | YES | BTREE | | | YES | NULL |
| orders | 1 | idx_orders_odate | 1 | odate | A | 2 | NULL | NULL | YES | BTREE | | | YES | NULL |
| orders | 1 | idx_orders_cnum | 1 | cnum | A | 7 | NULL | NULL | YES | BTREE | | | YES | NULL |
| orders | 1 | idx_orders_snum | 1 | snum | A | 5 | NULL | NULL | YES | BTREE | | | YES | NULL |
| orders | 1 | idx_orders_customer_date | 1 | cnum | A | 7 | NULL | NULL | YES | BTREE | | | YES | NULL |
| orders | 1 | idx_orders_customer_date | 2 | odate | A | 7 | NULL | NULL | YES | BTREE | | | YES | NULL |
| orders | 1 | idx_odate | 1 | odate | A | 2 | NULL | NULL | YES | BTREE | | | YES | NULL |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
7 rows in set (0.04 sec)
```

3. Create an index that would permit salesperson to retrieve his orders.

```
W2_99546_Pratiksha>CREATE UNIQUE INDEX idx_snum
-> ON ORDERS
-> WHERE SNUM = 1001;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'WHERE SNUM = 1001' at line 3
W2_99546_Pratiksha>SELECT * FROM ORDERS
-> WHERE SNUM=1001;

+-----+-----+-----+-----+-----+
| onum | amt   | odate   | cnum | snum |
+-----+-----+-----+-----+-----+
| 3003 | 767.19 | 1990-10-03 | 2001 | 1001 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+-----+-----+-----+-----+-----+
3 rows in set (0.02 sec)
```

4. Let us assume that each salesperson is to have only one customer of a given rating, and this is currently the case. Create an index that enforces it.

```
W2_99546_Pratiksha>CREATE UNIQUE INDEX IDX_SNUM_RATING
-> ON CUSTOMERS(SNUM, RATING);
ERROR 1062 (23000): Duplicate entry '1001-100' for key 'customers.IDX_SNUM_RATING'
W2_99546_Pratiksha>CREATE INDEX IDX_SNUM_RATING ON CUSTOMERS(SNUM, RATING);
Query OK, 0 rows affected, 1 warning (0.05 sec)
Records: 0 Duplicates: 0 Warnings: 1

W2_99546_Pratiksha>SHOW INDEX FROM CUSTOMERS;

+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| Table | Non_unique | Key_name | Seq_in_index | Column_name | Collation | Cardinality | Sub_part | Packed | Null | Index_type | Comment | Index_comment | Visible | Expression |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| customers | 1 | idx_salesperson_rating | 1 | snum | A | 5 | NULL | NULL | YES | BTREE | | | YES | NULL |
| customers | 1 | idx_salesperson_rating | 2 | rating | A | 6 | NULL | NULL | YES | BTREE | | | YES | NULL |
| customers | 1 | IDX_SNUM_RATING | 1 | snum | A | 5 | NULL | NULL | YES | BTREE | | | YES | NULL |
| customers | 1 | IDX_SNUM_RATING | 2 | rating | A | 6 | NULL | NULL | YES | BTREE | | | YES | NULL |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
4 rows in set (0.03 sec)
```

5. Create an index to speed up searching order on a given date by given customer.

```
W2_99546_Pratiksha>CREATE INDEX idx_odate_cnum
-> ON ORDERS(ODATE,CNUM);
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near ')ODATE,CNUM)' at line 2
W2_99546_Pratiksha>CREATE INDEX idx_odate_cnum
-> ON ORDERS (ODATE, CNUM);
Query OK, 0 rows affected (0.05 sec)
Records: 0 Duplicates: 0 Warnings: 0

W2_99546_Pratiksha>SELECT * FROM ORDERS
-> WHERE ODATE = '1990-10-03'
-> AND CNUM=2001;
+-----+-----+-----+-----+-----+
| onum | amt | odate | cnum | snum |
+-----+-----+-----+-----+-----+
| 3003 | 767.19 | 1990-10-03 | 2001 | 1001 |
+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```